

CUSTOMER DIRECT DELIVERY (CDD) DCT

1. OVERVIEW:

What is this use case about?

Answer:

Customer Direct Delivery (CDD) enables direct shipment of goods from suppliers to customers, bypassing Bosch warehouses to reduce handling, inventory, and logistics costs. It provides end-to-end visibility of CDD orders, tracking order status, goods in transit, invoicing, and delivery confirmation to ensure faster fulfillment, lower operational overhead, and improved customer service.

When is the report updated?

Answer:

Refresh - Hourly, 0:00

2. USER DOCUMENTATION:

This section outlines the step-by-step operational process of a Bosch CDD order, from customer placement to final delivery and invoicing.



2.1 CDD Process Flow

The CDD process is a streamlined sequence of actions ensuring efficient order fulfillment. Below are the detailed steps:

2.1.1 Open Sales Order (Customer Order Placement)

This is the initial stage where a customer places an order for a Bosch product. This order is then captured within Bosch's sales system (e.g., SAP).

2.1.2 Credit Check

After an open sales order is created, a mandatory credit check assesses the customer's financial viability and payment history, minimizing Bosch's financial risk.

2.1.3 Pending Purchase Order (PO Creation)

Post-credit check and payment, LOP2 initiates a PO to the external supplier, instructing direct shipment to the customer.

2.1.4 Open Goods Receipt Note (GRN) / Goods In Transit (GIT)

Post-shipment, GRN for CDD signifies "Goods In Transit" (GIT) directly to the customer, requiring shipment tracking.

2.1.5 Pending for Invoice

Upon customer receipt, the supplier invoices Bosch, and Bosch invoices the customer. This involves matching supplier invoices with purchase and sales orders.

2.1.6 Pending for LR (Lorry Receipt)

The LR is the transporter's document confirming goods receipt for transportation and serving as customer Proof of Delivery (POD). This final step confirms successful delivery and records the LR number.

2.2 Order & Cost Analysis

2.2.1 Customer Order Pattern

Shows: Billed Quantity (bar)

Total Value (secondary axis)

Customer count reference

Year-wise view (e.g., 2026)

Purpose: Provides visibility of billed quantity and corresponding value for customers in CDD.

2.2.2 Supplier Fulfilment Pattern

Shows: Sum of Billed Quantity (bar)

Total Value (secondary axis)

Customer code reference

Year-wise aggregation

Purpose: Provides visibility of fulfilment quantity and value under the CDD model.

2.2.3 Cost Benefit Tracking

Shows: Transportation Saving (monthly)

Warehouse Saving (monthly)

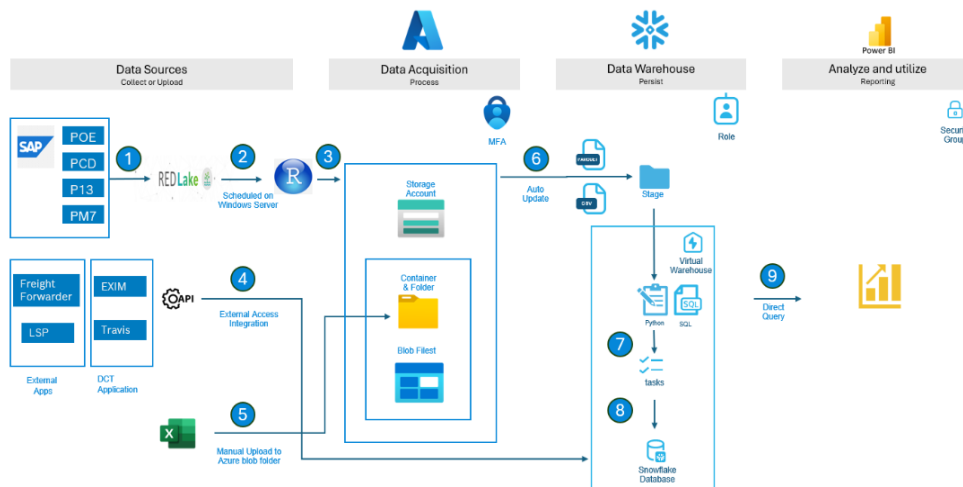
Cumulative Savings (Cum_Savings) trend (line)

Purpose: Provides visibility of monthly and cumulative savings generated through CDD execution.

3. TECHNICAL DOCUMENTATION:

This section briefly outlines the technical architecture underpinning the Bosch CDD system, highlighting its data flow and key components.

Overseas transportation architecture



3.1 CDD Technical Architecture Overview

The CDD system leverages a robust, cloud-enabled architecture to ensure reliable data acquisition, processing, storage, and reporting. It builds upon established Bosch data infrastructure to deliver actionable insights.

3.2 Key Architectural Phases

3.2.1 Data Sources:

SAP (POE, PCD, PL7): Core enterprise data from various SAP modules, covering procurement, commercial documents, and material management.

External Apps (Freight Forwarder, LSP, Travis, DCT App) & Excel: Integrates data from logistics partners, dedicated CDD applications, and allows for manual supplementary data uploads.

3.2.2 Data Acquisition:

RedLake & Windows Server: Extracts data from SAP via RedLake, with processes scheduled on Windows Servers for automation.

APIs & MFA: External application data is ingested via secure APIs. Multi-Factor Authentication (MFA) ensures secure access during data transfer.

Azure Storage Account: Data is stored in Azure Blob Storage (containers & folders) after initial acquisition, including manual uploads.

Auto Update: Automated processes ensure continuous data refresh within the storage account.

3.2.3 Data Warehouse:

Snowflake (Virtual Warehouse): Data is processed and persistently stored in Snowflake, a cloud data warehousing platform.

Staging: A temporary area for data validation and transformation.

Python & SQL: Used for complex data transformations, cleaning, and orchestration of data pipelines (tasks) within Snowflake.

3.2.4 Analyze & Utilize:

Power BI: The primary reporting tool, connecting directly to Snowflake via Direct Query. This enables near real-time dashboards and reports.

Security Groups: Access to Power BI reports is managed through security groups, ensuring data confidentiality and controlled access for authorized users.

This architecture ensures that the CDD system is scalable, secure, and provides timely, accurate data for proactive decision-making.

-> Power BI Dashboard Link: [Customer Direct delivery \(CDD\) - Power BI](#)

-> Snowflake Default Role Setup: [Snowflake Default Role Setup - 3S IT - End User Guide - Docupedia](#)

-> How to get Access: [DCT Access.docx](#)

-> Apply for the required role: [oneIDM](#)

QUESTIONS + ANSWERS

1. What is Customer Direct Delivery (CDD)?

ANSWER:

Customer Direct Delivery (CDD) is a supply chain execution model where goods are shipped directly from suppliers to customers without passing through Bosch warehouses.

Business Purpose:

- Reduce warehouse handling
- Lower inventory carrying cost
- Reduce logistics cost
- Improve delivery speed
- Provide end-to-end order visibility

CDD tracks:

- order status
- goods in transit
- invoicing
- delivery confirmation

2. What is the purpose of the CDD DCT dashboard?

ANSWER:

The CDD DCT dashboard provides operational visibility into Customer Direct Delivery order execution from customer order creation until final delivery and invoicing.

Business Purpose:

- Track CDD order lifecycle
- Monitor supplier fulfillment
- Track Goods In Transit (GIT)

- Monitor invoicing and delivery confirmation
- Analyze transportation and warehouse savings

The dashboard supports operational monitoring and execution control for direct delivery processes.

3. How does the CDD process work?

ANSWER:

CDD process flow:

1. Customer places sales order
2. Credit check is performed
3. LOP2 creates Purchase Order to supplier
4. Supplier ships goods directly to customer
5. Goods move under GIT tracking
6. Supplier invoices Bosch
7. Bosch invoices customer
8. LR/POD confirms final delivery

Final Outcome:

- Direct customer delivery without warehouse handling
- Faster fulfillment and reduced operational cost

4. What is Open Sales Order in CDD?

ANSWER:

Open Sales Order is the initial stage of the CDD process where a customer places an order for a Bosch product.

Operational Purpose:

- Capture customer demand in the sales system
- Initiate downstream execution activities

- Trigger credit and procurement processes

The order is recorded in Bosch sales systems such as SAP.

5. Why is Credit Check important in CDD?

ANSWER:

Credit Check validates the customer's financial viability and payment history before order execution.

Business Purpose:

- Reduce financial risk
- Prevent delivery to high-risk customers
- Ensure payment reliability before supplier execution

The credit check is mandatory before Purchase Order creation.

6. What is Pending PO in the CDD process?

ANSWER:

Pending PO refers to the Purchase Order creation stage in the CDD process.

Operational Flow:

- After successful credit check and payment validation
- LOP2 creates a Purchase Order to the external supplier
- Supplier is instructed to ship directly to the customer

This step initiates supplier-side execution for direct delivery.

7. What is GIT in the CDD process?

ANSWER:

GIT (Goods In Transit) in CDD represents materials shipped by suppliers directly to customers but not yet fully delivery-confirmed.

Business Purpose:

- Track shipment movement
- Monitor delivery status
- Maintain execution visibility during transportation

CDD GRN status indicates goods are in transit directly to the customer.

8. What is Pending Invoice in CDD?

ANSWER:

Pending Invoice stage occurs after customer receipt of materials.

Operational Flow:

- Supplier invoices Bosch
- Bosch invoices customer
- Supplier invoices are matched with Purchase Orders and Sales Orders

Business Purpose:

- Ensure financial reconciliation
- Validate execution completion
- Support billing accuracy

9. What is LR in the CDD process?

ANSWER:

LR (Lorry Receipt) is the transport document confirming goods receipt for transportation and delivery.

Business Purpose:

- Act as Proof of Delivery (POD)
- Confirm successful customer delivery
- Record transportation confirmation details

The LR number is recorded after successful delivery completion.

10. What is the role of LOP2 in the CDD process?

ANSWER:

LOP2 is responsible for initiating the Purchase Order to the external supplier after successful credit check and payment validation.

Operational Responsibility:

- Create supplier PO
- Trigger supplier execution
- Enable direct shipment to customer

LOP2 acts as the operational execution coordinator within the CDD process.

11. Why is CDD operationally beneficial?

ANSWER:

CDD improves operational efficiency by eliminating warehouse handling between supplier and customer.

Business Benefits:

- Lower inventory carrying cost
- Reduced warehouse dependency
- Lower transportation and handling cost
- Faster customer fulfillment
- Improved supply chain responsiveness

CDD also provides visibility into order execution and shipment tracking.

12. What is Customer Order Pattern analysis in CDD?

ANSWER:

Customer Order Pattern analysis provides visibility into billed quantity and total order value for customers under the CDD model.

The analysis includes:

- billed quantity
- total value
- customer count reference
- year-wise visibility

Business Purpose:

- Monitor customer demand trends
- Analyze CDD execution volume
- Support operational planning

13. What is Supplier Fulfilment Pattern analysis in CDD?

ANSWER:

Supplier Fulfilment Pattern analysis tracks supplier-side fulfillment quantity and value under the CDD model.

The analysis includes:

- billed quantity
- total value
- customer code reference
- year-wise aggregation

Business Purpose:

- Monitor supplier execution performance
- Analyze fulfillment trends
- Improve supplier coordination visibility

14. What is Cost Benefit Tracking in CDD?

ANSWER:

Cost Benefit Tracking measures operational savings generated through Customer Direct Delivery execution.

Tracked Metrics:

- Transportation Saving (monthly)
- Warehouse Saving (monthly)
- Cumulative Savings trend

Business Purpose:

- Measure financial benefit of CDD
- Track operational cost reduction
- Evaluate direct delivery efficiency

15. How frequently is the CDD dashboard refreshed?

ANSWER:

The Customer Direct Delivery (CDD) dashboard is refreshed hourly starting from 0:00.

This enables near real-time visibility into:

- order execution
- shipment tracking
- invoicing
- delivery confirmation

16. What are the main data sources used in the CDD system?

ANSWER:

The CDD system integrates operational data from multiple enterprise and logistics sources.

Primary data sources:

- SAP systems (POE, PCD, PL7)
- Freight Forwarder applications

- LSP systems
- Travis
- DCT App
- Excel uploads

These sources support procurement, logistics, shipment tracking, and reporting processes.

17. What is the role of Snowflake in the CDD architecture?

ANSWER:

Snowflake serves as the cloud data warehouse platform for the CDD system.

Business Purpose:

- Store and process operational execution data
- Support data transformation and orchestration
- Enable scalable reporting and analytics

The architecture uses:

- staging layers
- Python and SQL transformations
- Snowflake Virtual Warehouse processing

18. What is the role of Power BI in the CDD system?

ANSWER:

Power BI is the primary reporting platform used for CDD operational dashboards and reporting.

Business Purpose:

- Enable near real-time execution visibility
- Monitor order and logistics operations
- Support operational decision-making

Power BI connects directly to Snowflake using Direct Query.

19. Why is GIT visibility important in CDD?

ANSWER:

GIT visibility is important because materials move directly from suppliers to customers without warehouse buffering.

Operational Importance:

- Track shipment status
- Detect delivery delays
- Improve customer delivery visibility
- Support execution follow-up

Without proper GIT monitoring, delivery risks and customer service issues may increase.

20. Why is LR/POD important in the CDD process?

ANSWER:

LR/POD confirms successful transportation and final customer delivery in the CDD process.

Operational Importance:

- Validate delivery completion
- Support invoice closure
- Maintain shipment traceability

- Provide transportation confirmation records

It acts as the final execution confirmation document.

21. Required Roles to Access DCT and Snowflake?

ANSWER:

- i. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_GS_DEVELOPER
- ii. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_MA_DEVELOPER